Semantic Analysis: From Information Extraction to Semantic Role Labeling

Information Extraction

- **Definition:** Information extraction is the process of turning unstructured information embedded in texts into structured information (example: relational databases)

Named Entity Recognition

- The first step is usually Named Entity Recognition (NER), which we talked about earlier in the course
  - Named entities often serve to define which entities appear in the text

Citing high fuel prices, **United Airlines** said Friday it has increased fares by $6 per round trip on flights to some cities also served by lowercost carriers. **American Airlines**, a unit of **AMR Corp.**, immediately matched the move, spokesman **Tim Wagner** said. **United**, a unit of **UAL Corp.**, said the increase took effect Thursday and applies to most routes where it competes against discount carriers, such as **Chicago** to **Dallas** and **Denver** to San Francisco.

Legend:
Organization
Person
Location

Co-reference Resolution

- A common second step is finding equivalence classes of mentions that refer to the same entity

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Relation Extraction

- Next on our list of tasks is to discern the relationships that exist among the detected entities

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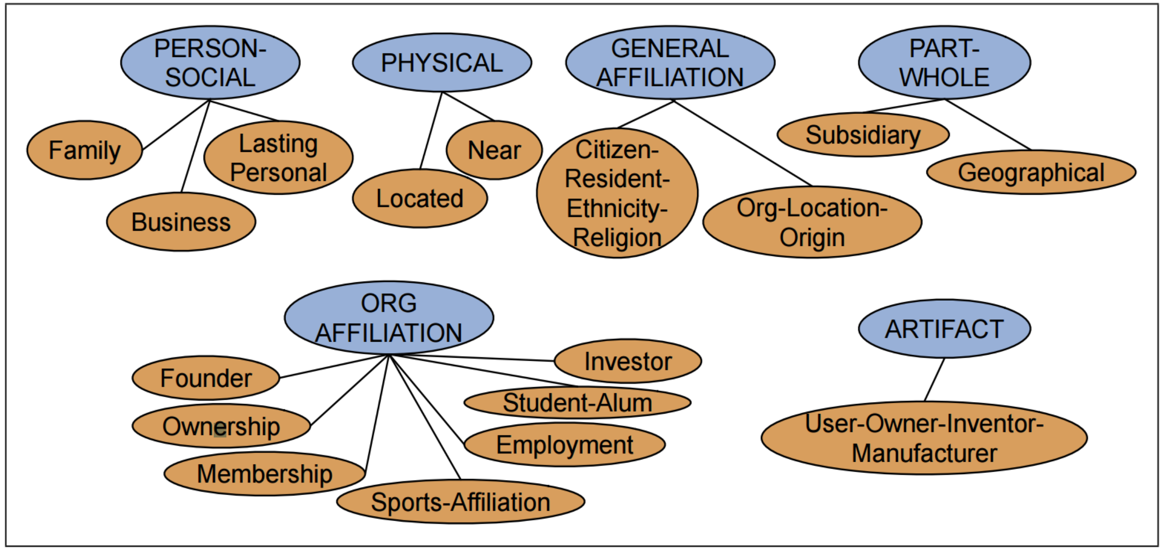
- For instance, the text tells us that Tim Wagner is a spokesman for American Airlines, that United is a unit of UAL Corp., and that American Airlines is a unit of AMR

Relation Extraction

- Often the relations are mapped to a pre-defined ontology
  - This is an example from the ACE shared task:



- For instance the aforementioned relations are instances of the PART-WHOLE relation:
  - “United is a unit of UAL Corp.”
  - “American is a unit of AMR”

ACE 2005 Multilingual Training Corpus
https://catalog.ldc.upenn.edu/ldc2006t06

Ontologies can be Mapped to Model-Theoretic Semantics

Domain	$\mathscr{D} = \{a,b,c,d,e,f,g,h,i\}$
United, UAL, American Airlines, AMR	a,b,c,d
Tim Wagner	e
Chicago, Dallas, Denver, and San Francisco	f,g,h,i
Classes	
United, UAL, American, and AMR are organizations	$Org = \{a,b,c,d\}$
Tim Wagner is a person	$Pers = \{e\}$
Chicago, Dallas, Denver, and San Francisco are places	$Loc = \{f,g,h,i\}$
Relations	
United is a unit of UAL	$PartOf = \{\langle a,b\rangle, \langle c,d\rangle\}$
American is a unit of AMR	
Tim Wagner works for American Airlines	$OrgAff = \{\langle c,e\rangle\}$
United serves Chicago, Dallas, Denver, and San Francisco	$Serves = \{\langle a,f\rangle, \langle a,g\rangle, \langle a,h\rangle, \langle a,i\rangle\}$

Relation Extraction

- There are many other ontologies of relations that have been defined
  - For example, UMLS, the Unified Medical Language System from the US National Library of Medicine has a network that defines 134 broad subject categories, such as:

Entity	**Relation**	**Entity**
Injury	disrupts	Physiological Function
Bodily Location	location-of	Biologic Function
Anatomical Structure	part-of	Organism
Pharmacologic Substance	causes	Pathological Function
Pharmacologic Substance	treats	Pathologic Function

Wikipedia Infoboxes

- Wikipedia also offers a range of relation types, drawn from the infoboxes
- For example, the Wikipedia infobox for Brad Pitt includes structured facts like *Occupation = "Actor"/"Producer"* or *Born = "Shawnee, Oklahoma"*
- These facts can be turned into relations like *born_in* or *occupied_as*
- These relations can complement relations extracted from text, or serve as features for them
- *Wikidata* (https://www.wikidata.org) is a knowledge base based on Wikipedia

Brad Pitt

Pitt at the premiere of *Fury* in Washington, D.C., October 2014

Born	William Bradley Pitt December 18, 1963 (age 53) Shawnee, Oklahoma, U.S.
Occupation	Actor • producer
Years active	1987–present
Works	Filmography
Home town	Springfield, Missouri
Spouse(s)	Jennifer Aniston (m. 2000; div. 2005) Angelina Jolie (m. 2014; separated 2016)
Children	6
Relatives	Douglas Pitt (brother)

Using Patterns to Discover Relations

- The earliest and still common approach for relation extraction is the use of lexico-syntactic patterns

NP {, NP}* {,} (and|or) other NP_H	temples, treasuries, and other important civic buildings
NP_H such as {NP,}* {(or|and)} NP	red algae such as Gelidium
such NP_H as {NP,}* {(or|and)} NP	such authors as Herrick, Goldsmith, and Shakespeare
NP_H {,} including {NP,}* {(or|and)} NP	common-law countries, including Canada and England
NP_H {,} especially {NP}* {(or|and)} NP	European countries, especially France, England, and Spain

Hearst, M. Automatic Acquisition of Hyponyms from Large Text Corpora, Proceedings of the Fourteenth International Conference on Computational Linguistics, 1992

Using Patterns to Discover Relations

- More modern approaches use additional features to define the patterns, such as named entity constraints
- For instance, if our goal is to answer questions about "who holds what office in which organization?", we can use patterns like the following:

PER, POSITION of ORG:
George Marshall, Secretary of State of the United States

PER (named|appointed|chose|etc.) PER Prep? POSITION
Truman appointed Marshall Secretary of State

PER [be]? (named|appointed|etc.) Prep? ORG POSITION
George Marshall was named US Secretary of State

Using Patterns to Discover Relations

- More accurate patterns still are ones that use syntactic information
- For instance, symmetric patterns can be used to discover synonymy or relatedness between entities

from **X** to **Y**

X and **Y**

X or **Y**

neither **X** nor **Y**

X as well as **Y**

Using Patterns to Discover Relations

- Syntactic information can filter a lot of noise
- What are the *Xs* and *Ys* in these cases:
  - "when they go to Austria, they like walking in the woods **as well as** skiing"
  - "apricots **and** other vegetables **and** fruit"
  - "Sandy is a Republican **and** proud of it"

from X to Y

X and Y

X or Y

neither X nor Y

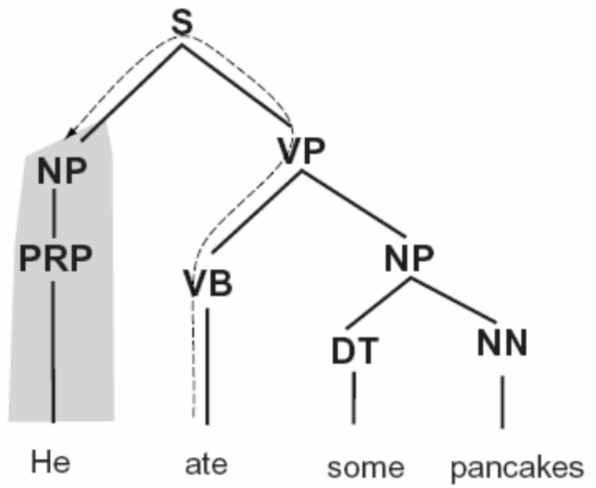
X as well as Y

Distant Supervision for Relation Extraction

- Hand-labeled data with relation labels is expensive to produce
- However, available resources such as Wikipedia infoboxes, have a great many relations that are in structured format
  - Wikipedia articles contain sentences that express these relations
  - This huge amount of examples allows us to define rich features

Path Features

- Path features:
  - Encode the path between two nodes, through the direction of the edges and their labels
- Similar features exist for dependency parses
  - We will see a bit of this in Exercise 5



Path	*Description*
VB↑VP↓PP	PP argument/adjunct
VB↑VP↑S↓NP	subject
VB↑VP↓NP	object
VB↑VP↑VP↑S↓NP	subject (embedded VP)
VB↑VP↓ADVP	adverbial adjunct
NN↑NP↑NP↓PP	prepositional complement of noun

Examples of Rich Features

...Hubble was born in Marshfield...
...Einstein, born 1879, Ulm...
...Hubble's birthplace in Marshfield...

PER was born in LOC
PER, born *, LOC
PER's birthplace is LOC

American Airlines, a unit of AMR, immediately matched the move, spokesman **Tim Wagner** said

Constituent path	$NP \uparrow NP \uparrow S \uparrow S \downarrow NP$
Base phrase path	$NP \rightarrow NP \rightarrow PP \rightarrow NP \rightarrow VP \rightarrow NP \rightarrow NP$
Typed-dependency path	$Airlines \leftarrow_{subj} matched \leftarrow_{comp} said \rightarrow_{subj} Wagner$

Schematized Algorithm for Distant Supervision

function DISTANT SUPERVISION(*Database D, Text T*) **returns** *relation classifier C*

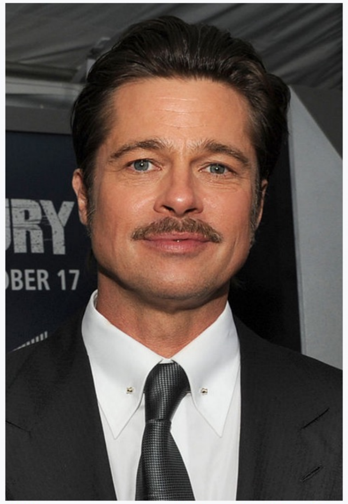
foreach relation *R*
  foreach tuple (*e1*,*e2*) of entities with relation *R* in *D*
    sentences ← Sentences in *T* that contain *e1* and *e2*
    f ← Frequent features in *sentences*
    observations ← observations + new training tuple (*e1*, *e2*, *f*, *R*)
C ← Train supervised classifier on *observations*
return *C*

Example: Distant Supervision

- From the infobox we learn that “Brad Pitt” and “Jennifer Aniston” were spouses
- The Wikipedia text states:
  - “In 2000, he married actress Jennifer Aniston; they divorced in 2005.”
  - “… playing a man with a grudge against Rachel Green, played by Jennifer Aniston, to whom Pitt was married at the time.”
  - “… Plan B Entertainment, a film production company he had founded two years earlier with Jennifer Aniston and Brad Grey”

- The first two examples are examples as to how spouse relation may be expressed in the text; the third example is noise

Brad Pitt



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Neural Information Extraction

- The simplest approach is to assume relations come from a closed set and to predict the relation given a sentence, where the subject and object are replaced with their NER tags
    - Assumes a binary relation

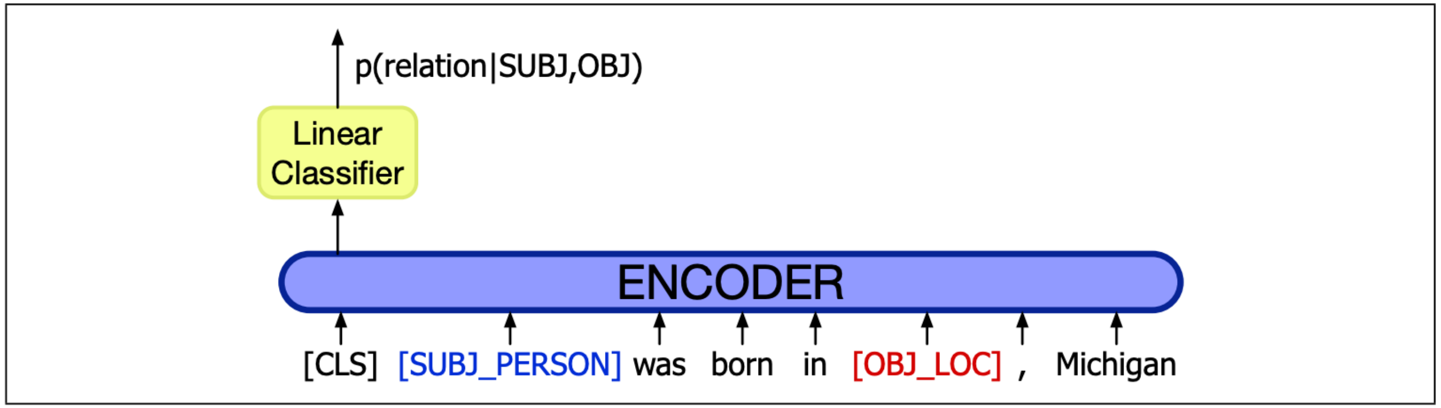


Figure 21.7 Relation extraction as a linear layer on top of an encoder (in this case BERT), with the subject and object entities replaced in the input by their NER tags (Zhang et al. 2017, Joshi et al. 2020).

https://web.stanford.edu/~jurafsky/slp3/21.pdf

Evaluation of Relation Extraction

- Semi-supervised methods are much more difficult to evaluate than supervised methods
  - They extract **new** relations from the web or a large text
  - As methods use very large amounts of text, it is impossible to test them in a sand box with a small pre-annotated gold standard
- Evaluation is therefore done by:
  - Computing precision
  - If the system can rank its output, we can measure precision for different output sizes (e.g., precision for 100 relations, 1000 relations etc.)
  - Extrinsic evaluation by testing how well these relations help, say, question answering

Poon and Domingos, *Unsupervised Semantic Parsing*, EMNLP 2009

Open Information Extraction

- Open Information Extraction (OpenIE) seeks to discover information from plain text
  - One way is to apply syntactic parsing and extract the relations it finds between named entities
  - This works well in some examples:

    "**Pitt** was born in **Shawnee, Oklahoma**"

    "**Jolie** gave birth to daughter **Shiloh Nouvel** in Swakopmund, Namibia, on May 27, 2006."

  - Extractions:

    born_in(Pitt, Shawnee)

    gave_birth_to_daughter(Jolie,Shiloh_Nouvel)

Open Information Extraction

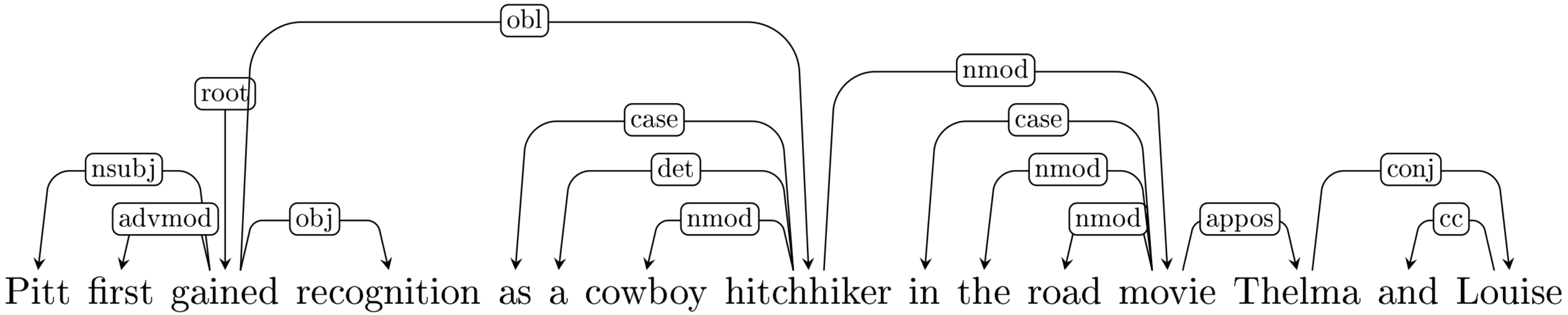
- Open Information Extraction (OpenIE) seeks to discover information from plain text
  - One way is to apply syntactic parsing and extract the relations it finds between named entities
  - Consider this example:

    Pitt first gained recognition as a cowboy hitchhiker in the road movie **Thelma and Louise** (1991). His first leading roles in big-budget productions came with the dramas **A River Runs Through It** (1992) and **Legends of the Fall** (1994), and **Interview with the Vampire** (1994).

  - What's the relation between "Pitt" and "Thelma and Louise"? Between "A River Runs Through It" and "Interview with the Vampire"?

Open Information Extraction

- The dependency path between *Pitt* and *Thelma and Louise* is *nsubj*↑*obl*↓*nmod*↓*appos*



- This is an indirect syntactic relation
  - It would be useful to have a representation that directly encodes the **semantic** relations between the participants

Semantic Role Labeling (SRL)

- The task of *Semantic Role Labeling* aims to represent a sentence in terms of its events or propositions
- Events consist of a main relation, participants and secondary relations

Powell met Zhu Rongji

Powell and Zhu Rongji met

Powell met with Zhu Rongji

Powell and Zhu Rongji had a meeting

. . .

battle

wrestle

join

debate

consult

Proposition: meet(Powell, Zhu Rongji)

meet(Somebody1, Somebody2)

Also others relation words, such as ‘meeting’, ‘summit’ etc.

When Powell met Zhu Rongji on Thursday they discussed the return of the spy plane.

meet(Powell, Zhu)   discuss([Powell, Zhu], return(X, plane))

FrameNet

- Organized by *frames*: a schematized event type
- A frame is invoked by frame elements, generally words or morphemes
  - They constitute the anchor of the scene and determine what it is about
- Semantic roles are defined per frames
- Example:
  - The *Judgment* frame
  - Frame elements: *admire, appreciate, value*…

Judge Evaluee Reason

She **blames** the government for failing to do enough to help

https://framenet.icsi.berkeley.edu/fndrupal